

LBDT: A Lightweight Blockchain-Based Data Trading Scheme in Internet of Vehicles Using Proof-of-Reputation

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Abstract—The exponential growth of data in the Internet of Vehicles (IoV) has created opportunities to improve traffic safety and efficiency through data trading. However, establishing trust among highly mobile and resource-constrained vehicles poses significant challenges for effective data trading in IoV. To address this issue, we propose a lightweight blockchain-based data trading scheme (LBDT), which ensures secure and efficient data trading in IoV. We introduce a proof-of-reputation (PoR) consensus mechanism to establish trustworthiness for data trading. Specifically, we use a progressive reputation mechanism to support the PoR consensus. LBDT utilizes a parallel-chain structure for the PoR consensus to minimize communication and storage costs while reducing transaction confirmation latency. Additionally, we adopt a double auction mechanism as an incentivizing strategy to encourage vehicle participation in data trading. We evaluate the performance of LBDT through extensive experiments. The experimental results demonstrate that LBDT is highly effective and secure, achieving a transaction latency of approximately 4 seconds. Moreover, LBDT successfully mitigates communication and storage overheads by over 90%, thus establishing its superiority over state-of-the-art solutions under comparable conditions.

Index Terms—Blockchain, data trading, Internet of Vehicles, trust management.

I. INTRODUCTION

THE integration of Internet of Things (IoT) with the automotive industry has resulted in a significant amount of data generated by onboard equipment in vehicles, which can be collected and shared to improve driving safety and quality. Vehicles can share real-time traffic information [1], [2], road conditions [3], [4], parking information [5], [6], and knowledge for federated learning [7], [8] through data trading. However, the high mobility nature of vehicles often complicates establishing

trust for effective data exchange. Therefore, ensuring trustworthy data trading plays a crucial role in IoV.

Previous studies on trust management of vehicles have primarily relied on a centralized rating mechanism [9], [10]. However, centralized systems are vulnerable to single points of failure (SPoF), where a failure in the central server can disrupt the entire network. In contrast, blockchain operates on a distributed network with replicated data across multiple nodes. This makes blockchain-based systems more reliable. Furthermore, blockchain provides a transparent and immutable ledger that ensures data integrity and traceability. Additionally, the decentralization in blockchain enhances autonomy and trust among participants in scenarios involving multiple stakeholders (e.g., vehicle manufacturers, service providers, and government agencies in smart city initiatives). Blockchain's decentralized consensus mechanisms validate transactions through multiple nodes, promoting transparency while reducing the risk of manipulation or fraud.

Recently, the integration of blockchain technology with IoV has been recognized as a promising solution for addressing trust and privacy concerns. Blockchain ensures consistency in distributed ledgers without relying on a trusted third party [11]. Consequently, it effectively solves problems faced by centralized schemes. Additionally, blockchain's anonymity and traceability ensure the privacy and security of data trading. However, due to vehicles' high mobility and resource constraints, blockchain-based data trading schemes in IoV need to address issues such as reducing transaction latency and minimizing communication and storage overheads. However, the implementation of blockchain poses challenges due to its complexity in processing, communication, and storage requirements. If vehicles were to function as full nodes, they would face issues with computing, communication and storage overheads. Therefore, it is crucial to devise a solution that allows vehicle participation in the blockchain network without overwhelming their constrained resources.

To address these issues, we propose a lightweight blockchain-based data trading scheme (LBDT) using parallel chains to facilitate secure and efficient data trading in IoV. We employ keyblocks and microblocks to decouple leader election from transaction serialization to reduce transaction latency and minimize storage and communication consumption. We assign lightweight roles to vehicles to address the issue of insufficient

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computing resources. Vehicles do not need to participate in the blockchain consensus as miners. Conversely, in light of the ample computational power and storage resources of Road Side Unit (RSU) [12], [13], RSUs are responsible for mining keyblocks that record reputation changes. Furthermore, RSUs with top reputations form a committee that utilizes Byzantine Fault Tolerance (BFT) [14] consensus to reach agreements on microblocks containing transactions. Additionally, we put forward a reputation-based double auction mechanism as an incentive for vehicle participation in data trading.

The main contributions of this paper include the following four aspects:

- We propose LBDT, a lightweight reputation-based consortium blockchain for ensuring secure and efficient data trading in IoV. By decoupling the lead selection from transaction serialization, LBDT effectively reduces transaction latency and communication overhead, thereby enhancing its adaptability to high mobility and limited resources in IoV.
- We present a reputation management mechanism that complements the reputation-based consensus, enabling progressive evaluation of nodes' reputations and identification of malicious nodes within the network.
- To promote honest data trading among vehicles, we conduct a reputation-based double auction mechanism.
- We conduct extensive experiments to assess the performance of LBDT, and our experimental findings validate LBDT's efficacy and efficiency. Moreover, LBDT exhibits superior scalability and shorter latency compared with state-of-the-art schemes.

The remainder of this paper is organized as follows. Section II introduces some related works. Section III presents preliminaries for LBDT design. Section IV presents models and problem formulation. Section V expounds on the design of LBDT. The evaluation is implemented in Section VII. Finally, the conclusion is made in Section IX.

II. RELATED WORKS

Several recent studies have investigated various applications of blockchain-based IoT [15], [16]. In the domain of energy trading, Li et al. [17] proposed a secure energy trading system by leveraging consortium blockchain. In the field of computation resources trading, Jiao et al. [18] presented a social welfare maximization auction in edge computing resource allocation to facilitate the utilization of blockchain in mobile IoT environments. In [19], Xiong et al. proposed a game theory-based resource management and pricing strategy for mobile blockchain to facilitate the offloading process.

In the area of data trading, Sadiq et al. [20] utilized consortium blockchain for secure data trading while avoiding duplication through hash lists accompanied by bloom filters for fast lookup. They also adopted an external distributed storage system, Inter Planetary File System (IPFS), to ensure the long-term availability of traded data. Chen et al. [21] proposed a blockchain-based IoV data trading scheme that utilizes an iterative double auction mechanism to maximize social welfare.

Zhao et al. [22] put forward a blockchain-based information sharing scheme with two chains: one for data uploading and the other for request uploading. Zheng et al. [23] introduced a hybrid architecture in their data sharing scheme, utilizing edge devices for distributed storage and incorporating a dual-chain structure to enhance efficiency and security. Behfar et al. [24] addressed the trade-off between on-chain/off-chain storage performance and security in data sharing based on blockchain through a dynamic or adaptive strategy. However, trust issues among participants, such as illegal data tampering, remain unresolved in the aforementioned scenarios.

Reputation models have been increasingly used in various domains over time. In the realm of blockchain, Zhuang et al. [25] proposed a consensus protocol that relies on reputation. The highest-reputation node acts as the leader for generating new blocks, and new blocks are validated and confirmed through reputation-based voting. Oliveira et al. [26] introduced a reputation-based consensus mechanism for blockchain, where nodes can append new blocks to the chain only if their reputation score exceeds a predefined network trust threshold. A group of randomly selected judges monitor consensus nodes' behavior and update their reputations. In the realm of double auctions, the work [27] proposed a reputation-based two-way auction mechanism that dynamically adjusts bids and offers based on participants' reputations, effectively encouraging genuine bidding even when both buyers and sellers may be dishonest. Reputation models are extensively employed in IoV as well. VRepChain [28], a social vehicle network system based on agreed-upon reputations by rating providers, addresses privacy concerns associated with unauthorized entities using ratings during operation. Additionally, Liu et al.'s work [29] presents an improved multi-source multi-weight subjective logic algorithm, which manages reputations integrating direct and indirect feedback from nodes using subjective logic trust models while considering factors such as event validity and familiarity.

To address the trust issues in IoV data trading, trust management and reputation model are introduced. Transactions and fraudulent behavior are permanently recorded on the ledger. Kang et al. [30] proposed a consortium blockchain for Vehicular Edge Computing (VEC) to establish a secure P2P data-sharing system while managing vehicle reputation using a three-weight subjective logic model. Zhou et al. [31] proposed a lightweight searchable data sharing scheme based on a reputation-based dynamic PBFT consensus mechanism, which combines public key encryption with equation testing and proxy re-encryption to achieve efficient data matching and sharing. Tong et al. [32] presented a traffic information interaction system based on non-subjective trust mechanism to solve bad-mouthing attacks. Liu et al. [29] proposed a multi-source multi-weighted subjective logic algorithm that integrates direct and indirect opinions of nodes. Huang et al. [33] proposed a data sharing scheme using Zero-Knowledge Proof (ZKP) to protect vehicle privacy while maintaining data auditability, along with a novel multi-sharding protocol to reduce communication overheads. However, these studies overlook the scalability and latency impact of their schemes, making them unsuitable for highly variable network topologies in the IoV domain. Consequently, we present LBDT,

a lightweight blockchain scheme for data trading in this paper. LBDT employs PoR consensus combined with a parallel-chain architecture to effectively minimize communication and storage overheads while reducing transaction confirmation latency.

III. PRELIMINARIES

In this section, we introduce the fundamental methods and technologies that serve as the basis for constructing the LBDT.

A. Blockchain

Blockchain is a decentralized, distributed ledger technology that records transactions across multiple nodes. The distributed nature of blockchain enhances reliability, while its cryptographic structure guarantees data integrity. Blockchain's transparency and decentralized consensus make it an ideal foundation for trust management and secure data trading in the highly dynamic IoV environment.

Consensus algorithms are crucial in blockchain systems as they enable distributed nodes to agree on the validity of transactions. Consensus algorithms can be divided into two categories: voting-based and proof-of-based. Practical Byzantine Fault Tolerance (PBFT) [34] is a typical consensus algorithm based on voting and is designed to tolerate Byzantine faults. It reaches consensus through multiple rounds of message exchange and voting among nodes. Proof-of-based algorithms, such as Proof-of-Work (PoW) [11] and Proof-of-Stake (PoS), are commonly employed consensus mechanisms. PoW is a widely adopted approach that necessitates nodes to solve puzzles to add new blocks. Despite its security, PoW is resource-intensive and introduces significant latency. In contrast, PoS is an energy-efficient alternative that determines block addition based on a node's stake in the network.

B. Double Auction

A double auction mechanism allows multiple buyers and sellers to trade goods simultaneously [27]. For example, most modern exchange markets (such as the New York Stock Exchange) use a double auction mechanism. In a typical double auction market, buyers submit bids (buy orders) to the auctioneer (market maker) indicating the highest price they are willing to pay for a certain good, while sellers submit offers (sell orders) setting the lowest price they are willing to accept to sell the good. The auctioneer collects orders and tries to match them by adopting certain market clearing policies to facilitate transactions.

C. Trust and Reputation

The concept of trust originates from social and psychological studies and is considered the foundation for long-term cooperation. Trust in IoV primarily focuses on the behavior of authenticated entities participating in communication. The high mobility and distributed nature of IoV pose significant challenges for designing trust models. One way to implement trust management in IoV is by recording and rating entity behavior, and calculating reputation.

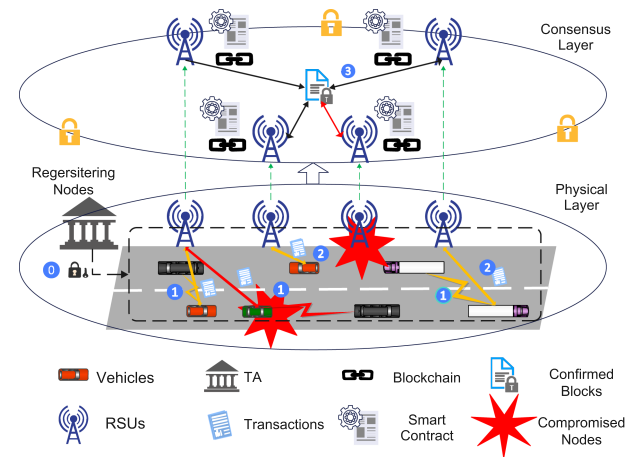


Fig. 1. System model of LBDT.

Since its introduction in 2010, the Gompertz function [35] has found wide applications in various fields, including biology, economics, and network analysis. The Gompertz function has well-suited characteristics for modeling processes that exhibit an initial phase of rapid growth followed by a slower, asymptotic approach to saturation. These characteristics align with the nature of reputation accumulation, where growth is limited by the number of successful interactions and user consistency over time. Therefore, the Gompertz function is commonly used to calculate reputation growth [36], [37], [38], reflecting gradual increases over time and rapid declines due to malicious activities.

IV. MODELS AND PROBLEM FORMULATION

A. System Model

The LBDT system, shown in Fig. 1, consists of two layers. The first layer is the physical layer, which includes vehicles and RSUs. Vehicles collect data from onboard devices and establish wireless communication with nearby vehicles and RSUs. Based on their behavior, vehicles can be classified as data sellers or data buyers. They upload their collected data to RSUs deployed along the road for timely services to nearby vehicles. Considering the abundant computational and storage resources, we designate RSUs as intermediaries responsible for matching and recording data trading between vehicles. Before joining the system, every node must register with a Trust Authority (TA), such as a government department, to obtain legitimacy. Additionally, TA handles key exchange tasks by generating public-private key pairs using the Elliptic Curve Digital Signature Algorithm (ECDSA) [39].

The consensus layer, represented by the second layer, addresses the variability in IoV and limited communication conditions of vehicles. Implementing a public chain on vehicles would be too costly. Therefore, we choose to deploy a consortium blockchain on RSUs. While both vehicles and RSUs serve as nodes in IoV, only RSUs can generate blocks. This blockchain acts as a distributed ledger that securely records interactions between nodes, ensuring trust within the IoV. By leveraging these recorded interactions, nodes can evaluate each other's reliability while reputation is calculated through aggregating opinions from diverse nodes.

TABLE I
NOTATIONS USED

Parameter	Definition
TX	Transactions
N	Number of committee members
f	Number of Byzantine members
Rep_i	The reputation value of node i
r	A random number
n	The interaction number
N_{peers}	The number of nodes with interactions
δ_i	The score of the i -th interaction
γ^{n-i}	The aging parameter of the i -th interaction
a, b, c	Parameters used to calculate the reputation
Z_k	The k -th region in the area
N_s, N_b	The number of sellers and the number of buyers
S_i, B_j	Data seller and data buyer
SO_i^k, BB_j^k	The bid from S_i and B_j in the region Z_k
SP_i^k, BP_j^k	The unit asking price and the unit bid price of data
SQ_i^k	The supply quantity and the demand quantity of data
t_i^k	The data collection time
SC_i^k, BV_j^k	The unit cost of data and the unit valuation of data
$\alpha_1, \alpha_2, \beta_1, \beta_2$	Weighting coefficients
p_{win}^k	The final trading price in the region Z_k
$T(i, j, k)$	The transaction between the buyer B_j and the seller S_i in the region Z_k
$\mathbf{W}_S$	The set of the winning sellers
$\mathbf{W}_B$	The set of the winning buyers
T_t, T_p, T_w	Transmission delay, propagation delay, and waiting time
N_{RSU}	The number of RSUs
N_V	The number of vehicles
$freq$	The data trading frequency
Stx	The size of a data transaction
S_{block}	The size of a block
B	The bandwidth of the network
R	The propagation rounds of the consensus algorithm
Δ	The upper bound of end-to-end delay

The runtime is divided into epochs by keyblocks, mined by RSUs, which contain changes in nodes' reputation from the previous epoch. At the start of each epoch, RSUs with top reputations are selected to form a committee for validating blocks through weighted voting. Subsequently, a leader is randomly chosen from this committee using a Verifiable Random Function (VRF) [40], who generates microblocks recording transactions uploaded by vehicles. Further details about this model are explained in Section V.

The second layer includes a smart contract that facilitates data matching between trading parties based on their requirements and supplies. Data sellers and buyers encapsulate their information as transactions, which are uploaded to nearby RSUs, triggering the smart contract to employ a double auction mechanism for requirement matching. Further details regarding this mechanism can be found in Section VI-C.

For a clear understanding, we list the notations of this paper in Table I.

B. Threat Model

The adversary can manipulate the system by delaying, dropping, re-ordering, inserting, or modifying messages at will. We assume that the cryptographic primitives used in this paper are secure and include a robust hash function and a reliable signature scheme.

Compromised RSUs: In real-world IoV scenarios, RSUs are often deployed without adequate security measures, making them vulnerable to attacks and compromise by malicious adversaries [41]. Once compromised, these RSUs can spread false reputation opinions or collude with other nodes to manipulate voting outcomes. However, we assume that adversaries cannot compromise more than 25% of the nodes due to prohibitive costs [42], and LBDT is designed to tolerate compromised RSUs possessing less than $\frac{1}{3}$ of the cumulative reputation of the entire committee.

Malicious Vehicles: The adversary can manipulate vehicles to launch attacks. We assume that malicious vehicles may deviate from the protocol in two ways: *i*) transmitting false information or dropping messages, such as transferring substandard data during data trading; and *ii*) tampering with reputation transactions to undermine other vehicles' reputation.

C. A Blockchain-Based Data Trading System for IoV

The workflow of the blockchain-based data trading system functions in the following steps.

Step 0: System initialization: During this stage, all network nodes register with the TA to obtain their key pairs and digital certificates. RSUs register as administration nodes by submitting user-specific information and making a supervised account deposit, which will be deducted in case of malicious behaviors [43], [44]. Due to limited resources and unstable connectivity, vehicles only install client applications without participating in consensus. All nodes are assigned a medium reputation value when they join the system, the growth rate of the reputation value decreases as it increases.

Step 1: Trading process: The trading process involves vehicles determining their roles based on requirements and resources. Data sellers register supply information with brokers (e.g., RSUs), while data buyers register demands. Supplies and demands are signed and encapsulated as supply transactions (TX_{sup}) and demand transactions (TX_{dem}). A transaction's format can be described as:

$$TX = [type || timestamp || information || Sig] \quad (1)$$

where *type* indicates the transaction type, *timestamp* represents the creation time, *Sig* is the node's signature of the transaction, and *information* depends on the content of the transaction. For example, in TX_{dem} , *information* includes data buyer's location, bid price, amount, and requirements. Brokers use a double auction mechanism to match demands and supplies (explained in Section VI-C).

Step 2: Reputation calculation and updating: Vehicles will be evaluated positively if they behave honestly in the deal, and the RSUs will be evaluated positively for collecting transactions and maintaining the consensus. On the contrary, entities will be

evaluated negatively for deviating from the protocol. After each interaction, the entity will encapsulate its feedbacks in reputation transactions TX_{rep} with its signature and broadcast it to the network. The RSUs will collect these reputation transactions and record them into blocks. The reputation of each entity can be calculated according to these opinions on the blockchain. More details about reputation calculation are described in Section V-B.

Step 3: Consensus process: The traditional consensus that combines transaction serialization with leader election compromises security and latency, while also posing challenges to limited vehicle resources in terms of storage and communication overhead. To address this issue without compromising security, we propose a reputation-based BFT consensus in a committee for microblocks, supplemented by a PoW consensus mechanism for keyblocks. Keyblocks record reputation changes in the previous epoch and become fixed after verification by the committee. Upon fixing a keyblock, a new epoch begins, and the committee is re-selected based on its reputation. The leader is randomly chosen from the committee and responsible for collecting transactions to generate microblocks. Detailed designs of the consensus will be discussed in Section V-A.

D. Problem Formulation

To establish a decentralized data trading system in IoV, it is imperative to implement a blockchain-based scheme that ensures security, scalability, and immediate transaction confirmation. Here we present the problem formulation of LBDT, encompassing crucial factors such as security, reduced latency, and improved scalability.

- **Security:** The security of the system is ensured when the cumulative reputation Rep_{adv} of the compromised committee members remains below $\frac{1}{3}$ of the aggregate reputations of all committee nodes. Mathematically, this condition can be expressed as $Rep_{adv} < \frac{1}{3} \sum_{i=1}^N Rep_i$, where N is the total number of committee members and Rep_i denotes the reputation of the i th node in the committee. Furthermore, the system must exhibit resilience against prominent attacks such as SPoF, double-spend attacks, and flash attacks.
- **Latency:** Latency refers to the time interval between transaction generation and its final confirmation on the blockchain. LBDT necessitates minimizing latency to ensure efficient and timely transaction processing.
- **Scalability:** The scalability refers to that LBDT should minimize communication and storage overheads for vehicles to efficiently handle an increasing number of vehicles, allowing for expansion without compromising performance.

V. PROTOCOL DESIGN

A. Reputation-Based Consensus Scheme

In the conventional blockchain system, peers solve cryptographic puzzles to reach a consensus [11]. However, due to limited resources and high mobility of vehicles, an appropriate choice for IoV applications would be a consortium blockchain with fixed identities and sufficient RSU resources. In LBDT,

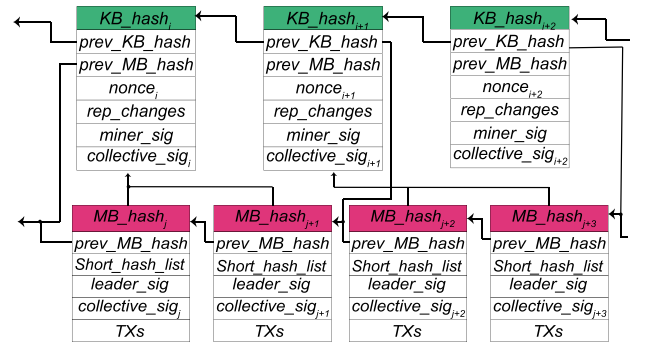


Fig. 2. Parallel-chain structure, which consists of a keychain (in green) and a microchain (in red).

RSUs function as administration nodes responsible for block generation and consensus while vehicles act as light nodes that submit transactions. To address vehicle topological variability and time-sensitive data in IoV, we adopt BFT consensus to reduce confirmation latency and introduce a reputation-based system to mitigate protocol attacks.

1) Blockchain Structure: In LBDT, we define two kinds of blocks (i.e., keyblocks and microblocks) like Bitcoin-NG [45] and ByzCoin [46]. Miners, i.e., RSUs, challenge the puzzles to mine the keyblocks. Unlike Bitcoin-NG, we decouple the miner of keyblocks and the leader of the microblocks. As shown in Fig. 2, we separate these two kinds of blocks into two parallel chains: keychain and microchain.

Microblock: The epoch is divided into rounds by microblocks. At the start of each round, a leader is elected from the consensus committee. This leader is responsible for aggregating and validating pending transactions, which are then encapsulated as microblocks. Each microblock contains a set of transactions and a hash value that references the previous microblock to ensure proper ordering. As shown in Fig. 2, an authorized microblock added to the blockchain also includes the collective signature of the committee and a hash value referencing the corresponding keyblock.

Keyblock: Keyblocks in LBDT serve two purposes: leader election and updating reputation abstracts. It contains reputation changes from the last epoch, calculated based on transactions in microblocks. Due to limited communication and storage resources, vehicles cannot download and store the entire microchain. Moreover, a significant portion of microchain transactions are unrelated to specific vehicles. Instead, the main focus for each vehicle is updating the reputation of other vehicles and RSUs. Therefore, by generating a keyblock for each epoch, LBDT provides a lightweight blockchain (i.e., keychain) that can adapt to communication and storage constraints of vehicles.

Similar to blocks in Bitcoin, the keyblock also incorporates a proof-of-work and a hash referencing the preceding keyblock: the proof-of-work is utilized for obtaining rewards, while the hash function ensures the order. However, unlike Bitcoin's progressive block confirmation, we employ BFT consensus to promptly validate the keyblock.

2) *Mining Keyblocks and Committee Selection*: The consensus of the committee is necessary for keyblocks and microblocks to be valid. Miners or leaders initiate a consensus instance for a block, which is then proposed to committee members for acceptance or rejection. To be considered valid, an instance must gather a quorum of signed votes. If there are f Byzantine nodes within the committee, the size of the committee N must exceed $3f + 1$. Otherwise, divergent decisions can be induced by Byzantine nodes in different parts of the system [47]. Therefore, it is crucial for the leader to collect at least $N - f \geq 2f + 1$ signed votes to determine the validation of a consensus instance.

Unlike the Nakamoto consensus [11], which requires the confirmation latency before a transaction can be confirmed (e.g., Bitcoin has a 1-hour confirmation latency), BFT consensus can reach an agreement immediately. However, the high communication complexity limits the committee size for BFT consensus. Therefore, for scalability reasons, we adopt Gosig [48] as the BFT consensus in LBDT due to its feasibility. Nevertheless, scaling up the committee to cover the entire system is still not practical. Hence, we need to design a membership determination scheme for the committee.

To form the committee, we use a reputation management mechanism based on historical behavior to determine membership. The committee consists of the most respected RSUs in the network and is re-selected at each epoch's beginning. The previous epoch concludes after confirming the keyblock. RSUs then join the committee according to their reputation ranking until the cumulative reputation reaches a target percentage of overall reputation.

3) *Committing Microblocks and Transactions*: The leader will be randomly selected from the committee using VRF once it is established. Each committee member can generate a verifiable random number r by utilizing its secret key and seed derived from the last keyblock, such as its hash value. Subsequently, they broadcast their respective verifiable random numbers r , and the nodes are designated as leaders for epochs in ascending order of r . The incorporation of a reputation mechanism outlined in Section V-B ensures that RSUs with higher reputations are less susceptible to corruption, enhancing the robustness against potential attacks.

The committee adopts a weighted-voting BFT consensus based on Gosig. Let $x_1, \dots, x_N$ represent the members of the consensus committee, and Rep_i denote the reputation value of node x_i . To make a consensus instance legal, the leader must collect signed aggregated votes [49] until the accumulated vote reputation reaches $\frac{2}{3}$ of the total committee reputation, which is $N_{vote} \geq 2f + 1 \wedge Rep_{vote} \geq \frac{2}{3} \sum_{i=1}^N Rep_i$. We will refer to this condition as the *legal threshold*.

As shown in Fig. 3, the leader of the current round packages pending transactions into microblocks B and then sends proposal $PR(B)$ to neighboring nodes in the gossip network to initiate a consensus instance. Other nodes verify proposals by checking the leader's signature, hash, and validating contained transactions within transaction set TXs . If a committee member accepts the proposal, it submits a signed abstract of the proposal $P(B)$ as a prepare message.

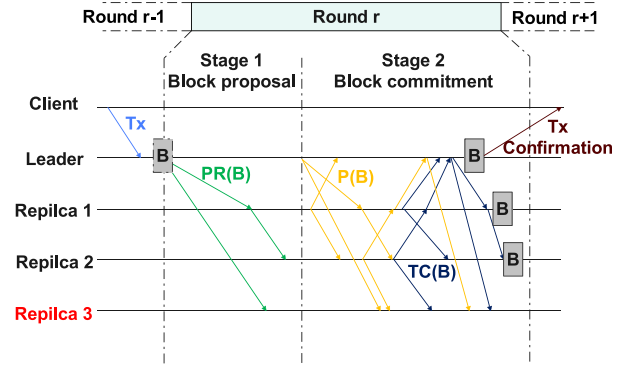


Fig. 3. Gosig consensus process, where red represents the adversary.

The nodes utilize the Gossip network to forward and store received $P(B)$. Aggregated signatures enable multiple $P(B)$ s to be combined into a certificate, denoted as $Cert_P(B)$. Once a node synthesizes a certificate with enough signatures surpassing the *legal threshold*, it enters the tentative-commitment phase. Similar to the preparation phase, the node generates and forwards the tentative-commitment message $TC(B)$. Likewise, when sufficient $TC(B)$ s are collected to generate a certificate, $Cert_TC(B)$, surpassing the *legal threshold*, the node enters the next epoch and appends the corresponding block to its local blockchain.

The committee starts each round with a timer. If no new microblock is generated before the timer expires, the current leader loses its eligibility to publish microblocks and it is transferred to the next committee member with the smallest VRF random number. Nodes can submit evidence of improper behavior or illegal transactions in a microblock as special transactions [45], [50]. Once the transaction is verified and appended to the blockchain, responsible nodes will face reputation loss and deposit deduction.

Additionally, Gosig utilizes the gossip network and compact block [51]. The PBFT consensus algorithm employs a clique communication method that involves point-to-point broadcasting for block propagation, resulting in high communication overhead. In contrast, Hotstuff [52] adopts a star-like communication method based on threshold signatures to reduce message complexity. However, this approach suffers from an unbalanced communication load as the leader's communication overhead is $O(n)$. To address this issue, Gosig leverages the gossip network to propagate blocks and transactions, thereby reducing the leader's communication overhead to a constant level. Moreover, Gosig incorporates block compression techniques to further minimize communication costs. Each node in Gosig maintains a pending transaction pool where new transactions are disseminated using the gossip network upon receipt. When a node becomes a leader, it forms a microblock by packaging from its own pending transaction pool. Instead of propagating the entire block directly, the leader only uses the gossip network to distribute the block head and an ordered list of short hashes. These six-byte short hashes correspond one-by-one with each

transaction contained within the block. Upon receiving a block proposal, nodes retrieve corresponding transactions from their pending transaction pools based on this short hash list and validate them accordingly. If any transaction is missing from their local pool, they can request other nodes to synchronize it. Through gossip network and compact block, Gosig can significantly reduce the confirmation delay of the consensus protocol, which will be discussed in Section VI-B.

B. Reputation Management Mechanism

In this paper, the reputation Rep_i of a node i is calculated based on its historical interactions. The reputation Rep_i increases when i demonstrates an honest behavior in the previous interaction, whereas it decreases if i engages in a dishonest conduct.

The reputation increases at a slow pace in reality. Once malicious interactions are detected, the reputation should decrease rapidly. To model this process accurately, we adopt the Gompertz function [35] to depict the growth of reputation. Furthermore, we introduce an aging function into the Gompertz function to give more weight to recent interactions. The aging function interaction can be expressed as:

$$r_n = \left(\sum_{i=1}^n \gamma^{n-i} \delta_i \right) \ln(N_{peers}) \quad (2)$$

$$\delta_i = \begin{cases} \delta_{pos}, & \text{if the interaction is positive} \\ \delta_{neg}, & \text{otherwise} \end{cases} \quad (3)$$

where n represents the interaction number, N_{peers} represents the number of nodes with interactions, δ_i is the score of the i th interaction, and γ^{n-i} is the corresponding aging parameter. $\gamma \in (0, 1)$ so that the recent interactions have more impact on the trust score. Moreover, to make it harder to build trust than to lose it, we set $\delta_{pos} < |\delta_{neg}|$ while $\delta_{pos} > 0$ and $\delta_{neg} < 0$.

The Gompertz function, which represents the reputation of Rep_i after the n th interactions, is based on the value of (2) and can be expressed as:

$$Rep_i(r_n) = a \times e^{-b \times e^{-c \times r_n}} \quad (4)$$

where a , b , and c are the function parameters. In this way, the reputation is limited from 0 to 1 and grows gradually, or drops rapidly.

We propose using blockchain as a global reputation ledger. Nodes will record interaction details as transactions and submit them. For example, after data trading between vehicles, buyers can leave negative feedbacks for sellers and attach the signed data of the sellers. Similarly, RSUs are evaluated based on their behaviors. Positive feedbacks are given for mining keyblocks while malicious actions like message delay and modification result in penalties. The leader verifies pending transactions and packages them into microblocks. Committee members review each transaction to ensure the legality and vote on the microblock. Only microblocks with majority approval will be added to the blockchain along with their valid reputation transactions.

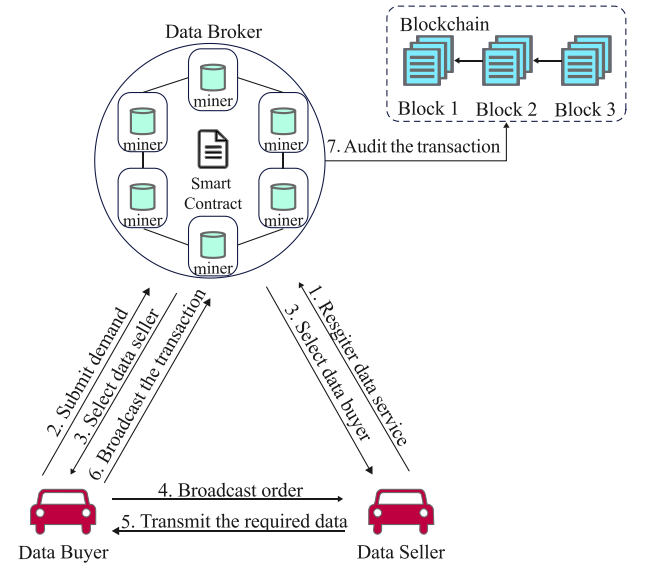


Fig. 4. Process of data trading through the utilization of the double auction mechanism.

Considering the majority of interactions are honest, we assume honesty by default. Grading for honest interactions does not require submission, and the corresponding node's reputation is updated using the default value. Conversely, if a node deviates from the protocol, its record of dishonest interaction will be submitted as a negative reputation transaction.

The RSUs on the microchain can calculate the reputation of other nodes. However, vehicles face challenges in performing reputation calculations. To address this, we propose keyblocks that capture reputation changes from the previous epoch and include hashes of microblocks for integrity. An RSU must incorporate a list of reputation modifications to generate a valid keyblock. For example, if node i starts with a reputation Rep_i^{n-1} at the beginning of the n th epoch, then after calculation, the updated reputation Rep_i^n will be recorded in the keyblock.

C. Double Auction Mechanism

Fig. 4 illustrates the process of data trading. Initially, data sellers register with brokers and provide details about their supplies (e.g., location, quantity, price). Subsequently, data buyers submit their demand information. The brokers collect this information and use a double auction mechanism to match supply with demand. Once matched, the buyer sends an order to the seller who verifies it before transmitting the signed data. After confirming the validity of the received order, the buyer broadcasts a payment transaction to the seller. These transactions are collected by brokers and serialized into microblocks.

In the case of a protocol deviation during data trading, such as the buyer's refusal to pay or the seller providing mismatched data, the other party can report this dishonest behavior through a reputation transaction with corresponding proof. Once confirmed by the committee, it will result in a rapid decline in the reputation. Additionally, we incorporate a deposit system. Once the reputation value falls below a certain threshold, the vehicle

will be banned from participating in market activities until it submits a new deposit.

We divide the area into different zones Z_k , where $k \in \{1, 2, 3, \dots, N_z\}$. Let N_s represent the number of data sellers and N_b denotes the number of data buyers. Sellers are denoted as $\mathbf{S} = \{S_1, S_2, S_3, \dots, S_{N_s}\}$, and buyers denoted as $\mathbf{B} = \{B_1, B_2, B_3, \dots, B_{N_b}\}$. The supply information from S_i in the region Z_k is represented as $SO_i^k = (SP_i^k, SQ_i^k, t_i^k)$, where SP_i^k is the unit asking price of the data from the zone Z_k , SQ_i^k stands for the corresponding quantity, and t_i^k represents the data collection time. The seller also has a corresponding unit cost $SC_i^k \leq SP_i^k$. When the seller does not have the data for the region Z_k , the corresponding SC_i^k and SP_i^k are 0. The supply information from S_i is sent as $\mathbf{Sup}_i = \{SO_i^1, SO_i^2, \dots, SO_i^{N_z}\}$ while the bid information from B_j is sent as $\mathbf{Bid}_j = \{BB_j^1, BB_j^2, \dots, BB_j^{N_z}\}$, where $BB_j^k = (BP_j^k, BQ_j^k)$ is its bid in the area Z_k with a unit price BP_j^k and the corresponding quantity BQ_j^k . The buyer has a corresponding unit valuation $BV_j^k \geq BP_j^k$. When the buyer does not have the data request for the Z_k zone, the corresponding BP_j^k and BV_j^k are 0. The detailed double auction mechanism is provided in Algorithm 1.

The process of the double auction algorithm can be summarized as following steps:

Step 1 (lines 1–5): Collecting information. The broker collects information from buyers and sellers. It groups this information by region and eliminates bids with a price of 0, then conducts a double auction in each group (lines 1–4). The broker first adjusts the seller's bid $\widehat{SP}_i^k = SP_i^k + \text{loss}(t_i^k, \text{Rep}_i)$ using a loss function (line 5):

$$\begin{aligned} \text{loss}(t_i^k, \text{Rep}_i) &= \alpha_1 \times \Delta t + \alpha_2 \times \Delta \text{Rep} \\ &= \alpha_1 \times (t_{\text{now}} - t_i^k)^{\beta_1} + \alpha_2 \times (1 - \text{Rep}_i)^{\beta_2} \end{aligned} \quad (5)$$

In (5), α_1 and α_2 represent weighting coefficients and $\alpha_1 + \alpha_2 = 1$, which are used to measure the importance of different attributes. We use Δt and ΔRep to represent the impact of data collection time and reputation on the transaction price, respectively. For example, we hope the data collection time is as recent as possible, so we can measure its impact by the difference between t_i^k and the current time t_{now} . t_i^k can also be replaced with one or more other data attributes. β_1 and β_2 are exponents that extend the model to a more general form.

Step 2 (lines 6–11): Searching for market clearing point. The broker sorts the offers from both the buyer and the seller (line 6–7). The seller's offers are sorted in ascending order of the adjusted price as $\widehat{SP}_{i,k}^1 \leq \widehat{SP}_{i,k}^2 \leq \widehat{SP}_{i,k}^3 \dots \leq \widehat{SP}_{i,k}^{N_s^k}$. The buyer's bids are sorted in descending order as $BB_{j,k}^1 \geq BB_{j,k}^2 \geq BB_{j,k}^3 \dots \geq BB_{j,k}^{N_b^k}$. As illustrated in Fig. 5, the offers of both buyers and sellers form a market clearing point, and the broker then finds the largest l and m that satisfy $\widehat{SP}_{i,k}^l \leq BP_{j,k}^m$ and $\widehat{SP}_{i,k}^{l+1} > BP_{j,k}^{m+1}$, where $\widehat{SP}_{i,k}^{l+1} = \infty$ if $l+1 > N_s^k$ and $BP_{j,k}^{m+1} = \infty$ if $m+1 > N_b^k$ (line 8). Let the final buying and selling

Algorithm 1: Reputation-Based Double Auction.

Input: The set of data sellers' offers $\mathbf{Sup}$, the set of data buyers' bids $\mathbf{Bid}$.
Output: The set of successful trades $\mathbf{T}$, the set of winning buyers $\mathbf{W}_b$, the set of winning sellers $\mathbf{W}_s$.
1: $\mathbf{T} \leftarrow \emptyset$, $\mathbf{W}_b \leftarrow \emptyset$, $\mathbf{W}_s \leftarrow \emptyset$;
2: **for** $k = 1$ **to** N_z **do**
3: Obtain the set $\mathbf{SO}_k = \{SO_i^k | SP_i^k > 0\}$ corresponding to Z_k from the offer set $\mathbf{Sup}$;
4: Obtain the set $\mathbf{BB}_k = \{BB_j^k | BP_j^k > 0\}$ corresponding to Z_k from the bid set $\mathbf{Bid}$;
5: Adjust the offer based on the attribute t_i^k and the reputation Rep_i according to (5):
 $\widehat{SP}_i^k = SP_i^k + \text{loss}(t_i^k, \text{Rep}_i)$;
6: Sort the $\mathbf{SO}_k$ in the ascending order of $\widehat{SP}_i^k$,
 $\widehat{SP}_{i,k}^1 \leq \widehat{SP}_{i,k}^2 \leq \widehat{SP}_{i,k}^3 \dots$;
7: Sort the $\mathbf{BB}_k$ in the descending order of $BB_{j,k}$,
 $BB_{j,k}^1 \geq BB_{j,k}^2 \geq BB_{j,k}^3 \dots$;
8: Find the largest l and m that satisfy $\widehat{SP}_{i,k}^l \leq BP_{j,k}^m$;
9: $p_{\text{win}}^k = \min(BP_{j,k}^m, \widehat{SP}_{i,k}^{l+1})$, where $\widehat{SP}_{i,k}^{l+1} = \infty$ if $l+1 > |\mathbf{SO}_k|$;
10: Remove the elements after $SO_{i,k}^{l-1}$ from $\mathbf{SO}_k$;
11: Remove the elements after $BB_{j,k}^{m-1}$ from $\mathbf{BB}_k$;
12: Sort both $\mathbf{SO}_k$ and $\mathbf{BB}_k$ in the descending order of the reputation;
13: $x = 0, y = 0$;
14: **while** $x \leq l-1$ or $y \leq m-1$ **do**
15: **if** $BQ_{j,y}^k < SQ_{i,x}^k$ **then**
16: $q_{ij} \leftarrow BQ_{j,y}^k$, $SQ_{i,x}^k \leftarrow SQ_{i,x}^k - BQ_{j,y}^k$, $y \leftarrow y + 1$;
17: **else if** $BQ_{j,y}^k > SQ_{i,x}^k$ **then**
18: $q_{ij} \leftarrow SQ_{i,x}^k$, $BQ_{j,y}^k \leftarrow BQ_{j,y}^k - SQ_{i,x}^k$, $x \leftarrow x + 1$;
19: **else**
20: $q_{ij} \leftarrow SQ_{i,x}^k$, $y \leftarrow y + 1$, $x \leftarrow x + 1$;
21: **end if**
22: $T(i, j, k) = (S_i, B_j, Z_k, p_{\text{win}}^k, q_{ij})$;
23: $\mathbf{W}_s = \mathbf{W}_s \cup S_i$, $\mathbf{W}_b = \mathbf{W}_b \cup B_j$.
24: $\mathbf{T} = \mathbf{T} \cup T(i, j, k)$;
25: **end while**
26: **end for**
27: **return** $\mathbf{T}$.

price be $p_{\text{win}}^k = \min(BP_{j,k}^m, \widehat{SP}_{i,k}^{l+1})$ (line 9). The first $l-1$ sellers and the first $m-1$ buyers win the auction and enter the matching stage (lines 10–11).

Step 3 (lines 12–27): Matching buyers and sellers. In the matching stage, both buyers and sellers are re-sorted in descending order of reputation (line 12). Sellers and buyers enter the matching one by one based on their reputations, and a trade $T(i, j, k) = (S_i, B_j, Z_k, p_{\text{win}}^k, q_{ij})$ is concluded, where q_{ij} is the smaller of supply $SQ_{i,x}^k$ and demand $BQ_{j,y}^k$. The traded volume q_{ij} reduces the supply and demand of both parties, while



Fig. 5. Seller and buyer curves in double auction market.

those with no remaining demand/supply complete the matching process and exit the auction; otherwise, they continue to wait for their next trading partner.

VI. SYSTEM ANALYSIS

A. Security

1) *Consensus Safety*: We focus on the safety of both transactions and the committee in relation to the PoR. Safety refers to consistency and liveness. PoW-like algorithms achieve probabilistic consistency through the longest chain principle. However, within a certain level of security, confirmation latency becomes interdependent with throughput [53]. As a result, PoW-like algorithms fail to ensure protocol security while accommodating rapid confirmation.

The proposed PoR utilizes Cosig for transaction confirmation. A leader suggests a proposal (such as a signed block with transactions), and all nodes vote on it to achieve a consensus (approval or rejection). We also introduce a weighted voting mechanism. To validate a proposal, the leader must gather enough signed votes surpassing the *legal threshold*. If the malicious node lacks sufficient reputation and numbers to exceed the *legal threshold*, there is no way to disrupt transaction consistency.

Transaction level liveness can be disrupted in two ways: (i) by preventing the committee from reaching a consensus, which will be discussed later if the number and reputation of malicious nodes exceed $\frac{1}{3}$ of the committee; (ii) by intentionally not publishing a microblock as a leader node, which can be addressed through implementing a timeout mechanism. At the beginning of each round, all nodes start a timer and if it expires without receiving any message from the leader, it triggers sending a *TimeOut* message. If signatures exceeding the *legal threshold* are received for *TimeOut* messages, it indicates that the agreement on timeout has been achieved within the committee leading to switching views for subsequent leaders.

We also need to ensure the safety of the committee: when the security assumptions mentioned in Section IV-B are met, (i) the number of malicious nodes within the committee cannot exceed $\frac{1}{3}$ of the committee or (ii) the reputation value of malicious nodes cannot exceed $\frac{1}{3}$. The safety of the committee can be modeled as a random sampling problem with two independent outcomes, namely honest nodes and malicious nodes. Each

TABLE II
COMMITTEE SAFETY

N	10	100	200	500	1000
$P[x < N/3]$	0.776	0.972	0.997	0.999	0.999
$P[x < 2N/3]$	0.996	0.999	1	1	1

sampling has a probability $p = 0.25$ of selecting a malicious node, and the number of samplings is the size of the committee N . Our objective is to calculate the probability that the number of selected malicious nodes is less than $N/3$. We employ the cumulative binomial distribution where x represents the count of malicious nodes selected into the committee:

$$P[x < N/3] = \sum_{i=0}^{N/3} \binom{N}{i} p^i (1-p)^{N-i} \quad (6)$$

Table II shows the results of (6) for different committee sizes N . The system guarantees both consistency and liveness when x is less than $N/3$. When X is less than $2N/3$, the system's liveness is compromised while consistency remains unaffected.

The probability of the system guaranteeing both consistency and liveness approaches 1 as the committee size N increases, as shown in Table II. Therefore, by appropriately selecting the committee size, we can ensure the safety of the committee.

Additionally, the reputation value accumulates over time, which reflects the cumulative contributions of RSUs. Unlike using PoW directly, PoR prevents adversaries from launching flash attacks through rented computing power. Moreover, if an adversary initiates an attack, its reputation will significantly diminish, requiring them to invest considerable time and effort into rebuilding it.

The utilization of PoW for keyblocks may result in forks when two nodes simultaneously complete the PoW. To address this issue, we employ the committee. When multiple keyblocks are generated concurrently, the committee arranges candidate blocks' head hashes in an ordered array and subsequently hashes them. The winning block is determined based on the remainder of the hash value modulo the number of candidate blocks.

Attackers may also attempt to manipulate reputation values to attack the system. Since the reputation is calculated based on behaviors recorded on the blockchain, manipulating this value necessitates falsifying records and obtaining the verification from the committee. For vehicles, forging records requires corresponding signature data; for RSUs, reputation values are primarily obtained through block mining, with mining records being securely stored on the blockchain. Neither of these records can be falsified. In order to manipulate reputations, attackers must first gain control over the committee. There is a condition for attackers to achieve this control: their combined number and reputation value must exceed $\frac{2}{3}$ of the total committee size; otherwise, they can only disrupt blockchain liveness without being able to alter blockchain records. Prior to gaining control over the committee, an attacker's ability to obtain the reputation is proportional to their computing power as they cannot directly manipulate the reputation. Assuming that attackers possess 25%

of the network's computing power, it becomes highly improbable for them to gain control when the committee size exceeds 100, as demonstrated in Table II.

2) *Resilience Against Specific Attacks: SPoF*: In centralized systems, all operations rely on a central server, creating a SPoF where any failure or attack can disrupt the entire system. In contrast, blockchain-based systems distribute data and transaction processing across multiple nodes, eliminating the risk of SPoF. However, even in a blockchain system, if the leader generation is predictable, an adaptive adversary can still cause SPoF through a denial of service (DDoS) attack [54]. For instance, in delegated Proof-of-Stake (DPoS) [55], the top 21 miners take turns generating blocks. Since these miners are relatively fixed (re-elected daily), an adaptive adversary can predict each round's leader and launch DDoS attacks against them. Therefore, LBBDT introduces randomness into leader elections using VRF to prevent adversaries from predicting leaders and launching attacks that result in SPoF.

Double Spending Attack: In a double spending attack [11], malicious nodes launch attacks by creating forks. Probabilistic consensus algorithms need a waiting period to ensure convergence of consistency. However, in the proposed PoR, the committee votes to determine keyblocks and microblocks. Therefore, the transaction confirmation delay can be reduced to less than a few seconds.

Selfish Mining Attack: In a selfish mining attack [56], the attackers hide their own blocks and mines alone, then release the blocks strategically to gain benefits and disrupt consensus. However, in the proposed PoR, the keyblocks and microblocks are immediately fixed. If the attacker hides their own blocks, he will only render the block invalid.

Eclipse Attack: In an eclipse attack [57], the attacker isolates and deceives the victim by controlling most of the links between the victim and the Bitcoin network. However, since the proposed PoR uses an aggregated signature, the attacker cannot forge a block or transaction even if he isolates the victim.

Flash Attack: In a flash attack [58], the proposed PoR prevents attackers from gaining a computational advantage through rented computing power. Even with substantial computing power, attackers still require significant time to accumulate reputation, which significantly reduces their ability to launch repeated attacks.

Bad-mouthing Attack: In terms of a bad-mouthing attack, the attacker provides fake transaction evaluations to influence the reputation adjustment results of other nodes [59]. In LBBDT, providing negative feedback requires presenting corresponding evidence, such as a buyer showing the data provided and signed by the seller. Leaders and validators will verify the evidence, such as checking if the seller's data deviates from its auction promise. A negative review must gather enough votes to be valid. Therefore, attackers cannot manipulate other nodes' reputations by falsifying negative reviews.

Undercover Attack: As to an undercover attack, the vehicle will pretend to be honest for a while to enhance its reputation and increase the chances of being selected as a consensus participant [60]. Our trust mechanism shows a rapid decline and slow recovery effect, implying that a dormant vehicle needs to

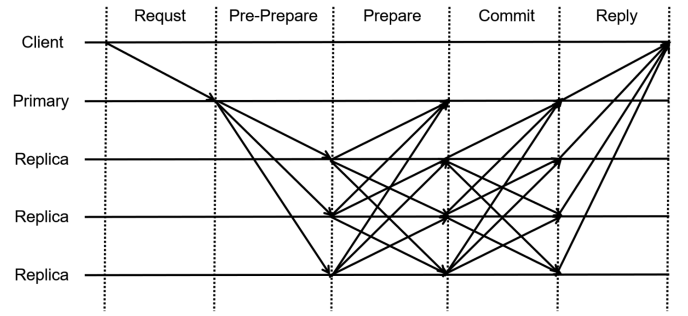


Fig. 6. PBFT.

engage in a prolonged honest behavior to become a participant, but it will lose its reputation and deposit if it launches an attack. The cost of such an attack is extremely high, outweighing the benefits. Therefore, rational vehicles would refrain from doing this.

Collusion Attack: In a collusion attack [61], participants in the network collaborate maliciously to manipulate system operations or outcomes. First, LBBDT employs BFT, which requires agreement from a supermajority of nodes to validate transactions, making it challenging for a colluding minority to influence consensus outcomes. Second, the feedback must include the cryptographic evidence of the transaction (e.g., signed data from the seller), which the committee uses to verify authenticity and disregard invalid feedback that does not impact reputations. Third, our reputation mechanism incorporates an aging function and considers the number of interaction parties, preventing attackers from quickly accumulating reputation through frequent transactions with controlled nodes. Lastly, we introduce factors like reputation into the auction mechanism to match buyers and sellers, which makes it difficult for attackers to predict auction results and reach deals with other attackers.

B. Latency

The confirmation latency of a transaction refers to the time elapsed from when the transaction is sent until it receives the final confirmation. The confirmation delay of a transaction can be quantified using the following formula:

$$\text{Latency} = T_{pend} + T_{trans} + T_{prop} + T_{proc} + T_{conf} \quad (7)$$

where T_{pend} denotes waiting time for pending transactions, T_{trans} is transmission delay, T_{prop} represents propagation delay, T_{proc} stands for processing time, and $T_{confirm}$ denotes waiting time for confirmation.

The $T_{trans} + T_{prop} + T_{proc}$ corresponds to a cycle of the consensus protocol. As shown in the Fig. 6, PBFT has a total of five phases of information transmission, with the first and last phases corresponding to the T_{pend} and T_{conf} , respectively, while the middle three phases correspond to a cycle of the PBFT protocol. The cycle of the protocol consists of multiple phases of information transmission, each of which involves message transmission, message propagation, and message processing.

The pending time refers to the time between when a transaction is sent and when it is included in a block proposal. Assuming

that all transactions proposed in a round can be included in the next round's block, the average pending time for a transaction is half of the consensus cycle.

The transmission delay refers to the time it takes for a node to receive and send data frames, which is dependent on the bandwidth B and the size of the data. The size of the data is determined by two factors, namely the size of the message S and the message complexity $Comp(N)$ related to the size of the committee N . For example, the message complexity of PBFT is $O(N^2)$, the HotStuff is $O(N)$, and Cosig is $O(C)$, where C is a constant. The transmission delay can be expressed as:

$$T_{trans} = \sum_{i=1}^R \frac{Comp_i(N) \times S_i}{B} \quad (8)$$

where R denotes the number of phases, $Comp_i(N)$ and S_i represent message complexity and message size of round i , respectively. T_{trans} is the sum of the transmission delay used in each phase. Assuming that the bandwidth of each node is the same, the transmission delay of each round depends on the node with the highest load. For example, in PBFT, the leader needs to send a block proposal to $N - 1$ nodes in the pre-prepare phase, which corresponds to a transmission delay of $((N - 1) \times S_{block})/B$.

As to the propagation delay, it is the time it takes for a message to travel end-to-end. If we denote Δ as the upper bound of the end-to-end delay for a message to propagate between any two nodes, then the propagation delay of a transaction depends on the number of propagation phases:

$$T_{prop} = \sum_{i=1}^R \Delta \quad (9)$$

The processing time T_{proc} refers to the time a node spends processing received messages. In LBDT, nodes need to verify the validity of received blocks and votes, such as checking the hash of the block header, the legality of transactions in the block, and the validity of signatures in the vote.

The waiting time T_{conf} is the time it takes for the client to confirm the execution result of a transaction. For example, in LBDT, vehicles can confirm the transaction result by downloading the corresponding microblock. In this case, the waiting time is the time it takes for vehicles to download the microblocks. T_{conf} includes three parts, which are the transmission time, the propagation time and the time for consistency convergence:

$$T_{conf} = \frac{S_{block}}{B} + \Delta + t_{conv} \quad (10)$$

where t_{conv} denotes the time required for consistency convergence, which can be up to an hour for PoW-like algorithms and is zero for BFT-like algorithms.

Using (7)–(10), we can derive (11) to represent the transaction latency:

$$\begin{aligned} Latency = 1.5 \times \sum_{i=1}^R \left(\frac{Comp_i(N) \times S_i}{B} + \Delta \right) \\ + \frac{S_{block}}{B} + \Delta + t_{conv} \end{aligned} \quad (11)$$

In general, the propagation delay and the processing time are relatively small and constant (Δ is typically less than a few hundred milliseconds). Among BFT-like algorithms, the main factor is the transmission delay. Assuming $N = 100$, $S_{block} = 1$ MB, and $B = 1$ Mbps, the transmission delay of the PBFT in the pre-prepare phase is approximately 800 s. In contrast, suppose that the leader in PoR has 5 adjacent nodes in the gossip network, and the sizes of the transaction and the short hash are 256 B and 6 B respectively, then the transmission delay of the PoR in the block-proposal stage is approximately 1 s as it uses the compact block and gossip network.

PoR also reduces communication and storage costs. Assuming a committee size of 100, a data trading frequency of $freq = 1$, an epoch time of 60 seconds ($T = 60$ s), a number of vehicles of 100000 ($N_V = 100,000$) and the reputation record size of 4 B, we can estimate that the keyblock's size would be around 40 KB. Assuming that transaction sizes are 100 B ($S_{tx} = 100$ B), the microblocks' accumulated size within one epoch will be $freq \times N_V \times T \times S_{tx} = 1$ MB, which is 25 times larger than the according keyblocks. The vehicle only needs to synchronize keyblocks when no related transaction is happening. As a result, LBDT not only reduces storage overhead but also minimizes communication overhead for vehicles, thereby improving the system's scalability.

C. Auction Analysis

A feasible auction mechanism should satisfy individual rationality, budget balance and truthfulness.

Theorem 1 (Individual Rationality): In a successful trading, the seller's final price is higher than its offer, while the buyer's final price is less than its bid, i.e., $\forall T(i, j, k) \in \mathbf{T}$, it satisfies $p_{win}^k \geq SP_i^k \wedge p_{win}^k \leq BP_j^k$.

Proof: For the case that S_i wins in the region Z_k , we have $p_{win}^k = \min(BP_{j,k}^m, SP_{i,k}^{l+1})$ while both $BP_{j,k}^m > SP_{i,k}^l$ and $SP_{i,k}^{l+1}$ is not less than $SP_{i,k}^{l-1}$, which is the largest offer among the winning sellers in the region Z_k according to Algorithm 1. Therefore, we have $p_{win}^k \geq SP_i^k$. Similarly, for a buyer $B_j \in \mathbf{W}_b$, we have $p_{win}^k \leq BP_{j,k}^m < BP_{j,k}^{m-1} \leq BB_j^k$. $\square$

Theorem 2 (Budget Balance): The auction-based incentive mechanism is a budget balancing mechanism if the auctioneer has no deficit. In other words, the total price charged to all winning buyers must be greater or equal to the total amount paid to all sellers:

$$\sum_{T(i,j,k) \in \mathbf{T}} P_s \geq \sum_{T(i,j,k) \in \mathbf{T}} P_b \quad (12)$$

Proof: According to the algorithm, the final prices for both the seller and the buyer are p_{win}^k . Therefore, the two-way auction algorithm is strongly Budget Balancing. $\square$

Before giving Theorem 3, we define the utility function of the seller S_i :

$$U_{i'}^s = \begin{cases} \sum_{T(i',j,k) \in \mathbf{T}} (p_{win}^k - SC_{i'}^k) \times q_{i',j}, & S_{i'} \in \mathbf{W}_S \\ 0, & \text{else} \end{cases} \quad (13)$$

and the utility function of the buyer $B_{j'}$:

$$U_{j'}^b = \begin{cases} \sum_{T(i,j',k) \in \mathbf{T}} (BV_{j'}^k - p_{win}^k) \times q_{i'j'}, & B_{j'} \in \mathbf{W}_B \\ 0, & \text{else} \end{cases} \quad (14)$$

Theorem 3 (Truthfulness): The utility obtained by the user's dishonest quotation cannot be higher than that obtained by its honest quotation (i.e., $SP_i^{k'} = SC_i^k$, $BP_j^{k'} = BV_j^k$).

Proof: For both the seller and the buyer, according to Algorithm 1, they cannot change their trading quantities by adjusting their bid information, so they can only try to obtain more unit utility by modifying their bids.

For a seller S_i , supposing that the honest quotation is SA_i^k , we need to prove that among all possible quotations $SP_i^{k'}$, $SP_i^{k'} = SC_i^k$ maximizes the utility.

Case 1: $SC_i^k + \text{loss}(t_i^k, \text{Rep}_i) < \widehat{SP}_{i,l,k}^l$. If the seller bids truthfully, it will win the auction; if the quotation is $SP_i^{k'} + \text{loss}(t_i^k, \text{Rep}_i) = \widehat{SP}_i^{k'} < \widehat{SP}_{i,l,k}^l$, then the buyer still belongs to $\mathbf{W}_S$ and has the same utility, as the winning price p_{win}^k is independent of $\widehat{SP}_i^{k'}$; if the quotation is $\widehat{SP}_i^{k'} \geq \widehat{SP}_{i,l,k}^l$, then $\widehat{SP}_i^{k'}$ becomes too high to win. For this situation, S_i gains zero utility for failing in the auction.

Case 2: $SC_i^k + \text{loss}(t_i^k, \text{Rep}_i) \geq \widehat{SP}_{i,l,k}^l$. If the bids truthfully, he will lose in the auction with a utility equal to 0; if he bids equal or more, then he will still lose in the auction; if he bids less so that $\widehat{SP}_i^{k'} < \widehat{SP}_{i,l,k}^l$, he will win in the auction with a negative utility. If the seller tries to modify data information such as t_i^k (the seller cannot modify the reputation value because it is recorded and maintained by the blockchain), then the buyer can file a lawsuit through the reputation transaction after receiving the data. Once the lawsuit is successful, the seller's deposit will be deducted, and its reputation value will be significantly reduced, damaging its long-term interests.

Similarly, for a buyer B_j , the utility obtained by the untruthful quotation cannot be higher than that obtained by the honest quotation. $\square$

VII. EXPERIMENTS

In this section, we evaluate the performance of LBBDT, as well as present the comparison with state-of-the-art methods.

A. Implementation

To assess the effectiveness of LBBDT, we develop a prototype of LBBDT using SUMO¹ version 0.32.0, integrated with OM-NetPP² version 5.4.1. The simulation is conducted in a region measuring 10 km $\times$ 10 km, as depicted in Fig. 7. The region consists of 400 RSUs and 1,000 vehicles, which are evenly distributed throughout the area. The vehicles move at a constant speed on the road network. The wireless communication is modeled using IEEE 802.11p. In addition, we set the data trading



Fig. 7. Simulation scenario map.

TABLE III
SIMULATION PARAMETERS

Parameters	Setting
Map size	10 Km*10 Km
Number of Vehicles	[10000,1280000]
Vehicles speed	15 m/s
Number of RSUs	[400,12800]
Coverage range of RSU	300 m
MAC model	IEEE 802.11p
Road type	Two-way
Channel bandwidth	10 MHz
Fading model	Nakagami fading
Transmission power	20 dbm
Data trading frequency	1/60 per second
Rate of compromised node	[10%,60%]
Transaction size	256 bytes
Keyblock interval	60 s
Microblock size	1 MB
Simulation time	3 days

frequency of the vehicles to once per minute on average (i.e., 1/60 per second), with each transaction size being 256 bytes. The average interval between keyblocks is 60 seconds, while microblocks have an average interval of 1 seconds. The experiments are conducted on Ubuntu 16.04 running on VMware Workstation hosted by a personal computer equipped with an Intel(R) Core(TM) i7-10750H CPU @2.60 GHz and supported by 32 GB RAM capacity. Table III presents the key parameters used in our experiments, where most of the parameters are adopted from previous works [62], [63].

B. Security

Here we analyze the security of LBBDT by comparing it with three top-performing protocols, i.e., the Multi-Weight Subjective Logic (MWSL) scheme [62], the PoW-based scheme proposed by Chen et al. [21] and the blockchain-based IoV with

¹[Online]. Available: <https://eclipse.dev/sumo/>

²[Online]. Available: <https://omnetpp.org/>

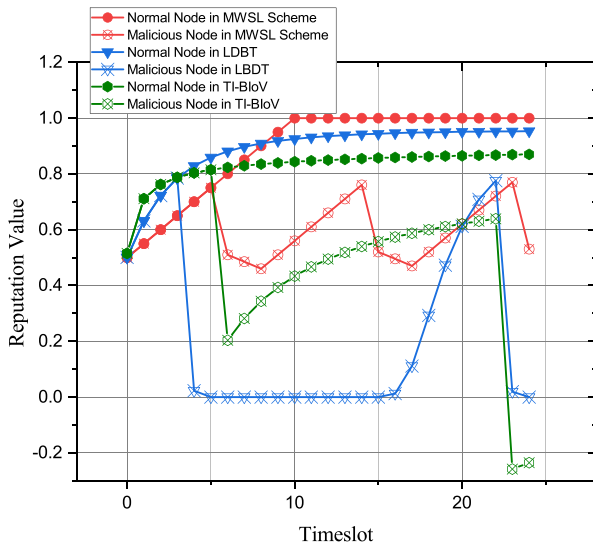


Fig. 8. Reputation in both normal and malicious scenarios.

trust and incentive (TI-BIoV) [32]. We conduct simulations for all protocols using the parameters specified in Section VII-A.

First, we compare the reputation model. Both MWSL and TI-BIoV utilize reputation mechanisms to record nodes' interactions. MWSL employs a linear model that integrates opinions from all participants to calculate a node's reputation. In contrast, TI-BIoV simulates the node's reputation value using an arctan function, where the growth of reputation for positive interactions diminishes over time, while the reduction of reputation remains constant and is significantly greater than the growth. Fig. 8 illustrates the dynamics of reputation for normal and malicious nodes across different schemes. Malicious nodes pretend to be honest to accumulate reputation before launching attacks. Generally, after an attack occurs, the reputation of malicious nodes experiences a rapid decline. Notably, in LBDT scheme, this decline is more pronounced and takes longer to recover from, thereby increasing the cost of attacks for adversaries. TI-BIoV also follows a similar approach but with decreasing values for reputational growth per interaction over time. This makes it challenging for newly joined nodes to accumulate sufficient reputation, thus compromising system fairness.

Then, we examine the compromised RSUs that deviate from the protocol. As elaborated in Section IV-B, these compromised RSUs have the potential to disrupt the system's consistency and liveness. Chen's mechanism utilizes the PoW consensus algorithm, relying on the longest chain principle for consistency. MWSL employs an enhanced DPoS algorithm, where the top 21 RSUs take turns producing blocks. To prevent collusion attacks among RSU nodes, MWSL sets a reputation threshold and stipulates that RSUs meeting the threshold become validators in proportion to system size. The consistency is ensured by an agreement from over two-thirds of validators. TI-BIoV uses a reputation-based PBFT, where the top 3 nodes take turns producing blocks. In Fig. 9, successful attacks and computing power controlled by adversaries are illustrated. In a small-scale system with 1,000 vehicles, the PoW-based scheme

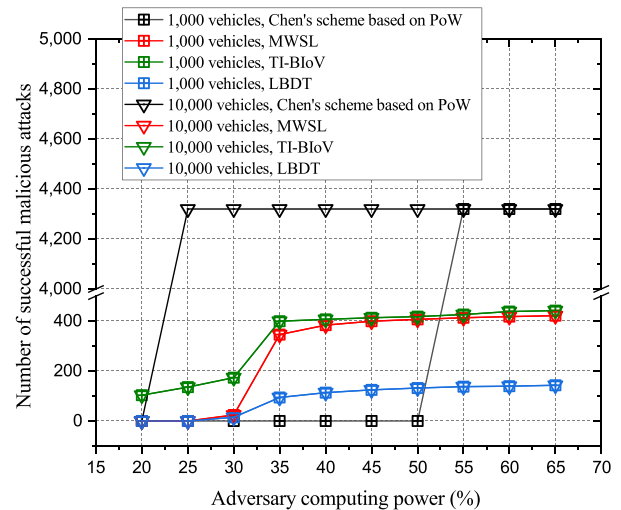


Fig. 9. Number of successful attacks launched by different adversaries.

demonstrates superior performance with 51% fault tolerance but becomes vulnerable once adversaries control more than 51% of the computing power. On the other hand, LBDT requires a longer recovery period for adversaries to accumulate reputation before launching subsequent attacks, limiting their frequency and increasing attack costs significantly. Furthermore, as system size increases, PoW-based schemes' security diminishes; when there are 10,000 vehicles in the system, their fault tolerance drops to only 25%. This expansion leads to larger block sizes and increases fork probabilities which enhance successful attack likelihoods for adversaries (Section VI-A). However, other schemes remain unaffected by system size due to their vote-based mechanisms. Moreover, in TI-BIoV since block proposers can be predicted (the three most reputable nodes), adversaries can launch DDoS attacks against these nodes to disrupt system liveness. In contrast, in LBDT, the leader is randomly generated based on VRF, making it difficult for adversaries to disrupt its liveness.

C. Latency

As analyzed in Section VI-B, the transaction latency depends on the consensus cycle and waiting time T_{conf} . Assuming the end-to-end delay is 100 ms, the node transmission load is the main factor affecting the delay.

The impact of block size on transaction latency is examined first. The relationship between block size and latency reflects the system's throughput. According to Section VI-B, for a blockchain system in IoV, the larger the block size per unit of time, the more vehicles it can accommodate. In PoW-like algorithms, transaction latency mainly depends on waiting time, while the block size affects security. For voting-based consensus algorithms, the block size directly influences transaction latency. MWSL and TI-BIoV generate a significant load during the proposal stage by sending blocks to all committee members, whereas LBDT only needs to send compact blocks to a constant number of nodes. Additionally, LBDT reduces the load of

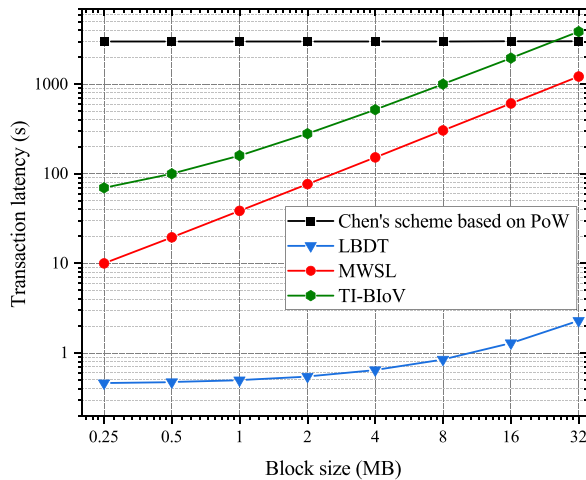


Fig. 10. Relationship between the transaction latency and the block size.

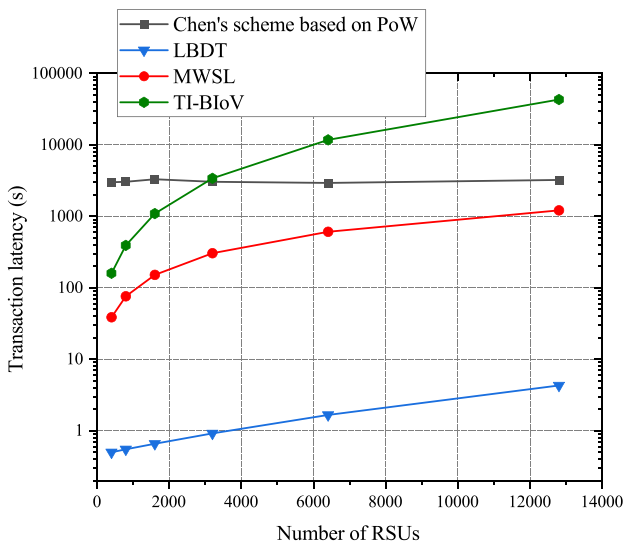


Fig. 11. Relationship between the transaction latency and the number of RSUs.

downloading transaction for vehicles via microblocks, thereby reducing waiting time. As shown in Fig. 10, when the block size is 0.25 MB, LBDT achieves a transaction latency of only 0.5 s compared to 3 s with a 32 MB block size for LBDT. Assuming each transaction has a size of 256 B, LBDT can complete approximately 60 000 transactions per second. According to the given transaction frequency $f = 1/60$ per second for each vehicle, LBDT can accommodate up to 36 000 000 vehicles. On the other hand, both MWSL and TI-BIoV experience rapid increases in delay with larger block sizes exceeding 1,000 s which is impractical for IoV.

The number of RSUs impacts the consensus cycle, as shown in Fig. 11. In the PoW-based mechanism, increasing the number of RSUs barely affects the transaction latency, which remains at approximately 1 h. Similarly, the committee size in the MWSL based on the enhanced DPoS varies proportionally with the number of RSUs, thereby resulting in a linear increase in transaction latency. However, for LBDT and TI-BIoV, an increase in RSUs

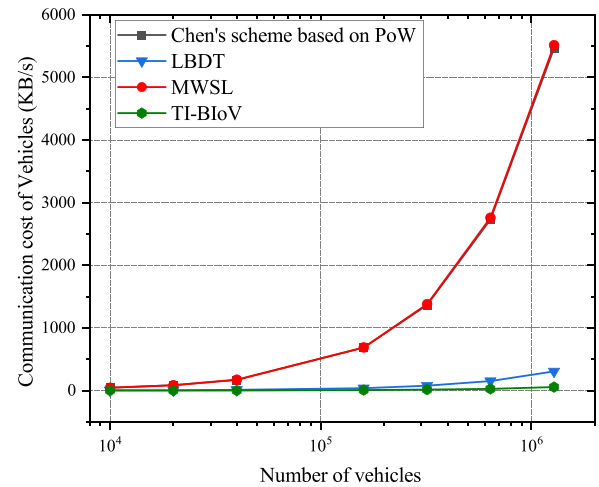


Fig. 12. Relationship between the communication overhead and the number of vehicles.

impacts the consensus committee's composition, consequently leading to an extended transaction latency. Despite this effect, our scheme's parallel chain structure maintains a transaction latency of approximately 4 seconds, representing only 10% compared to that of MWSL.

D. Scalability

We first explore the scalability of the three schemes for IoV concerning communication overhead and number of vehicles, and the results are illustrated in Fig. 12. The vehicle communication overhead refers to the consumption of communication resources by each vehicle for synchronization blocks per unit of time. The communication overhead is primarily used for synchronizing transaction information, and in schemes that incorporate reputation management mechanisms, it also includes the synchronization of the reputation update information. As shown in Fig. 12, LBDT exhibits significantly lower communication costs compared to MWSL and the PoW-based scheme, as these schemes require nodes to download the whole blockchain. LBDT only requires the synchronization of a fraction of transaction information due to the parallel chain structure employed. For instance, when using a keyblock epoch of 60 s and a microblock round of 6 s and assuming the vehicle is only related to one transaction during this epoch, the vehicles only need to download the whole microblock containing its transaction information and the heads of other microblocks and the keyblock. Similarly, TI-BIoV significantly reduces the communication overhead of vehicles by using SPV schemes. However, SPV has a serious problem that the validation of transactions depends on connecting to multiple full nodes (RSUs) to avoid eclipse attacks. This is more likely to happen in communication-limited vehicular networks. In contrast, even if a vehicle in the LBDT is attacked by eclipse attacks, the attacker cannot forge the committee's aggregated signature to disrupt consistency.

Subsequently, we investigate the scalability of the three schemes by storage overhead and simulation time, where the

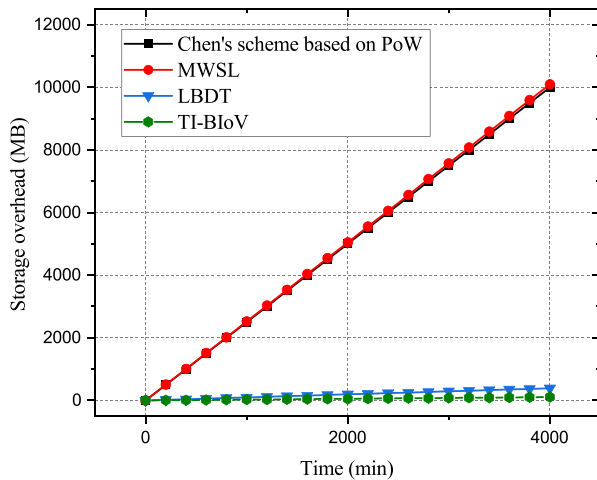


Fig. 13. Relationship between the storage overhead and the simulation time.

number of vehicles is set to 10000, as shown in Fig. 13. Similar to the communication overhead, when the vehicle needs to store the whole blockchain, the storage overhead will increase quickly and linearly with time. However, in LBDT, the vehicle only needs to retain the keyblocks. Therefore, LBDT can effectively reduce the storage overhead and achieve better scalability. Upon reaching a system scale of 10000 vehicles and conducting 4,000 minutes of experimentation, both the PoW-based scheme and MWSL exhibit storage overhead exceeding 10000 MBs. In contrast, LBDT demonstrates a storage overhead below 200 MB under identical conditions.

In summary, similar to that of latency, LBDT surpasses state-of-the-art competitors concerning scalability.

VIII. DISCUSSION AND FUTURE WORK

LBDT has shown promising results in addressing the issue of IoV data trading. However, there are still some potential limitations in future research. First, this paper does not propose customized privacy protection mechanisms. Privacy protection is of great importance in vehicle-to-vehicle data trading. For example, users may be concerned about their trading records being leaked and being inferred to reveal their daily movement trajectory. The lack of customized privacy protection mechanisms can be solved by combining existing privacy protection solutions. Such as using Conditional Privacy-Preserving Authentication (CPPA) [64], [65], [66] directly. The most direct approach is to have the vehicle preload a large number of public/private key pairs and corresponding certificates from TA onto its onboard device [67]. The vehicle can randomly select a pair of public/private keys for signature verification, achieving anonymous authentication and hiding the vehicle's true identity.

Another issue is that although LBDT has a good scalability, its transaction latency increases exponentially as the number of RSUs increases linearly. Although our experiments show that LBDT can maintain low delay and outperform state-of-the-art solutions in scenarios with more than 10000 RSUs, which covers most real-world scenarios such as the RSU system under

construction in Beijing,³ the growing transaction latency and the future possibility of even larger IoV still pose challenges. A feasible solution is to introduce the sharding mechanism. Each shard corresponds to a fixed physical region, such as a district. A sub-committee is generated within each shard, and all sub-committees together elect a committee. The committee is responsible for coordinating transactions between shards. Since the size of each shard is smaller, the transaction latency can also be controlled. Through the sharding mechanism, the scalability of LBDT can be further improved to adapt to larger IoV scenarios.

IX. CONCLUSION

In this paper, we proposed LBDT, a blockchain-based data trading scheme for IoV that addresses the challenge of insufficient trust by leveraging PoR and parallel chains. LBDT combines PoW and BFT based on a trust management mechanism, which greatly reduces the communication overhead and delay of transactions while ensuring security, making it well-suited for IoV environments with high topological variability and limited resources. Furthermore, we introduce a double auction mechanism to incentivize vehicle participation in data trading. Experimental involving up to 10000 nodes demonstrate that LBDT is highly effective and secure, achieving a transaction latency of approximately 4 seconds. Moreover, LBDT successfully mitigates communication and storage overheads by over 90%, thus establishing its superiority over state-of-the-art solutions under comparable conditions.

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